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Name:Jessrel James Galagate **Date: June 3, 2026**

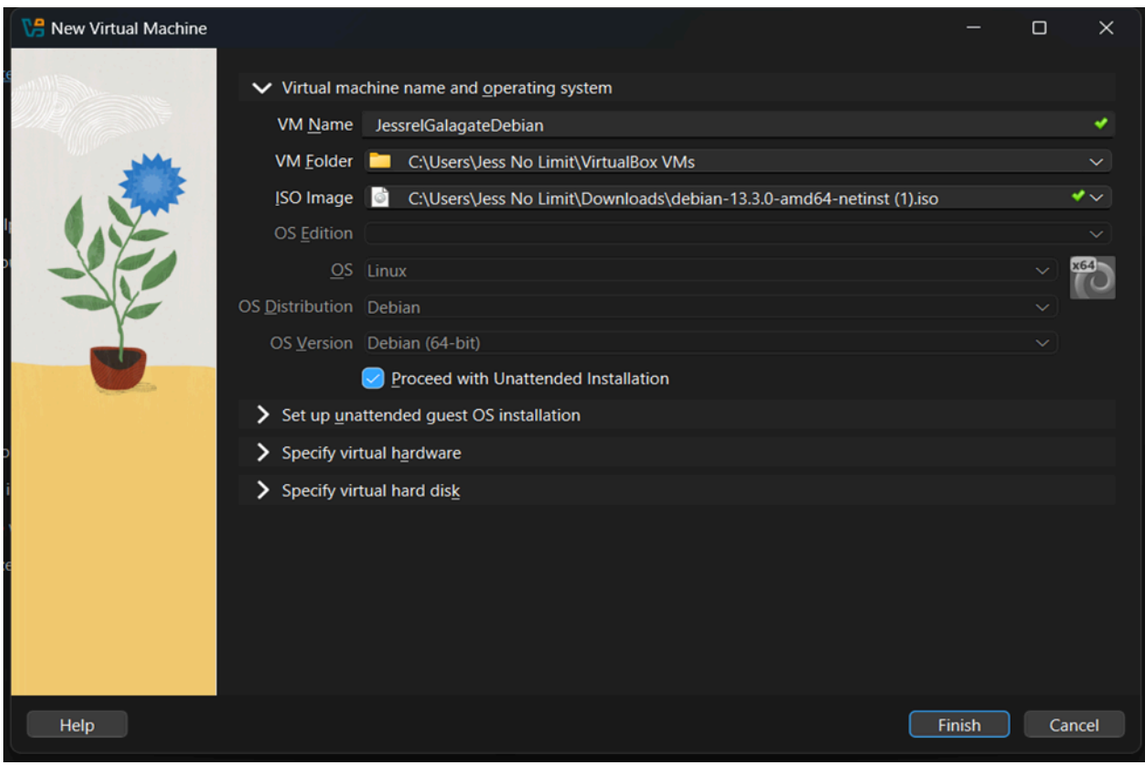
Course: IT 112-IRTG3

Laboratory Exercise No.3 Linux Installation (Debian) and User, Group, and File Permission Management in Linux

Linux Installation (Debian)

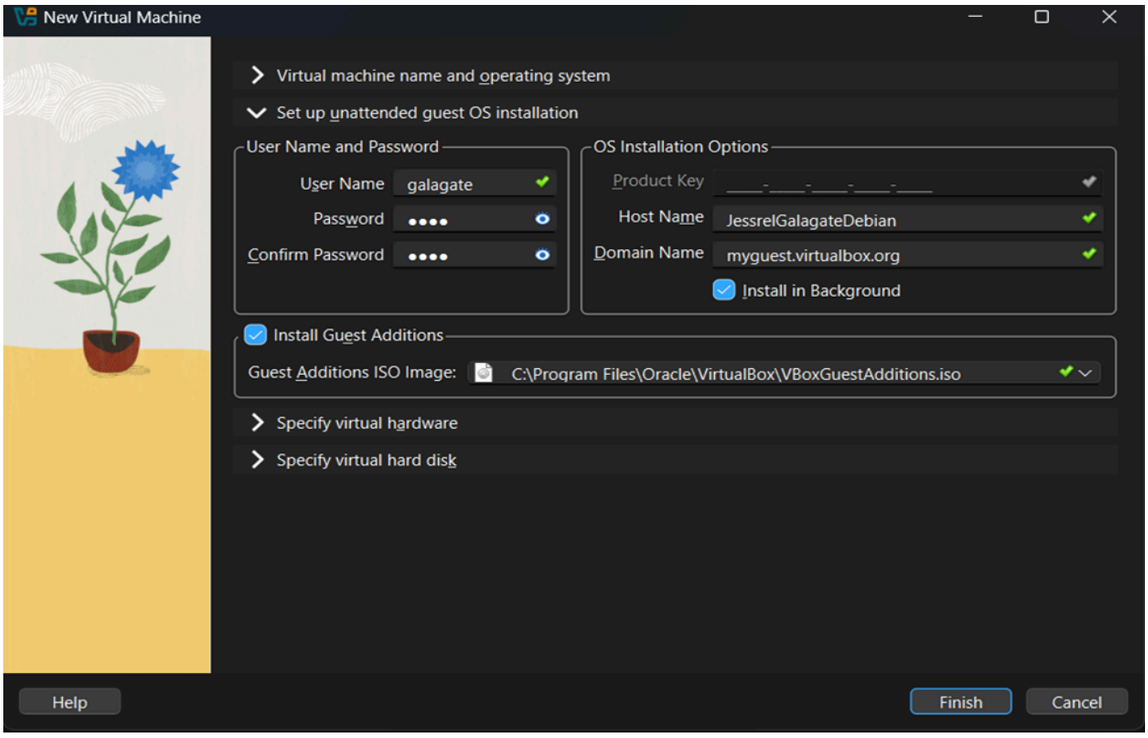
1.1 Virtual Machine Creation

- Open Virtual Box
- Click New
- The virtual machine name was set **EX: Debian Linux VM**
- The Debian ISO file was selected as the bootable installation media (64bit).



1.2 Set up unattended guest OS installation

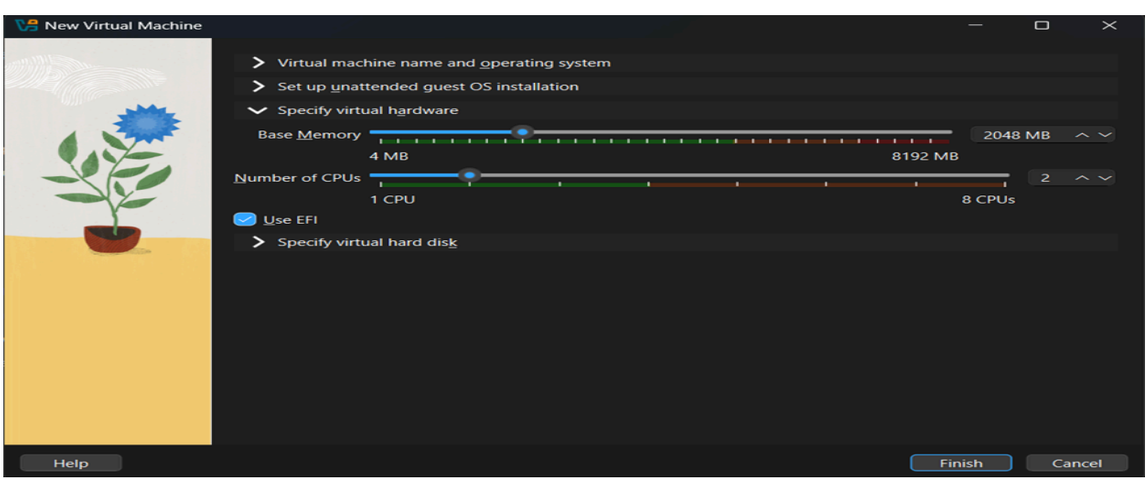
- Set User name and Password
- Set Host Name



1.3 CPU and Memory Allocation

- **System resources were manually assigned during VM configuration:**
  - **Base Memory (RAM): 2048 MB (2 GB)**
  - **Processor (CPU): 2 CPU cores**
  - **Firmware: EFI enabled**

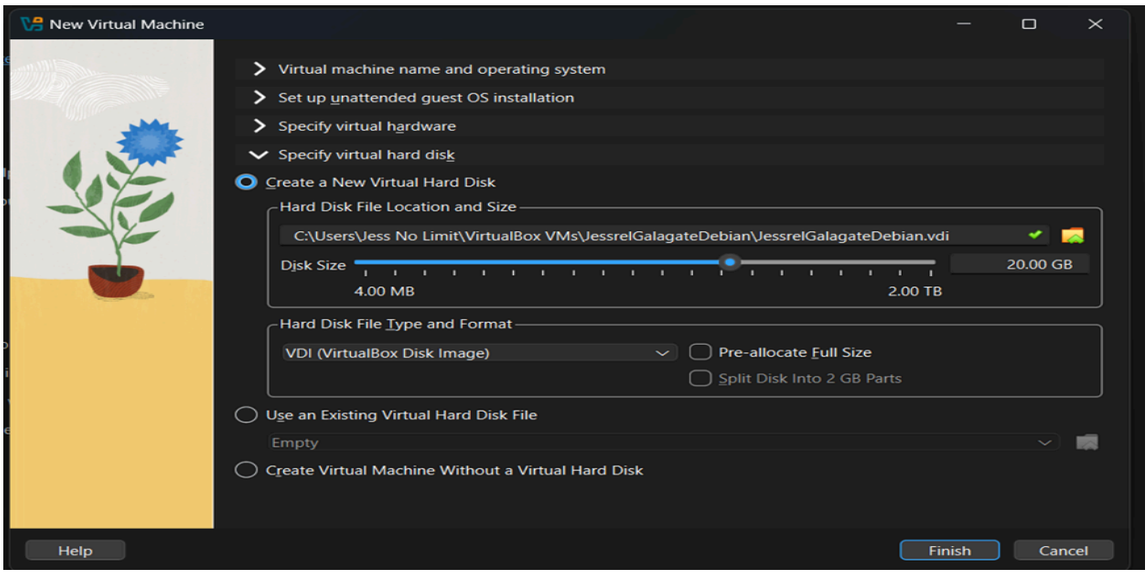
These values were selected to provide sufficient performance for basic system operations and operating system installation while maintaining efficient use of the host machine's hardware resources.



1.4 Virtual Hard Disk Configuration

- A new virtual hard disk was created during the VM setup process:
  - Hard Disk Type: VDI (VirtualBox Disk Image)
  - Disk Size: 20.00 GB
  - Allocation Method: Dynamically allocated (Pre-allocate Full Size unchecked)
  - Disk File Location: Stored in the default VirtualBox VMs directory

These settings were chosen to provide sufficient storage for the operating system and essential applications while efficiently using host system disk space through dynamic allocation.



1.5 Debian Installation Process

- Select Debian/Linux
- Loading may be a bit longer but it is normal



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System Information and Commands

2.0 System Information Collection

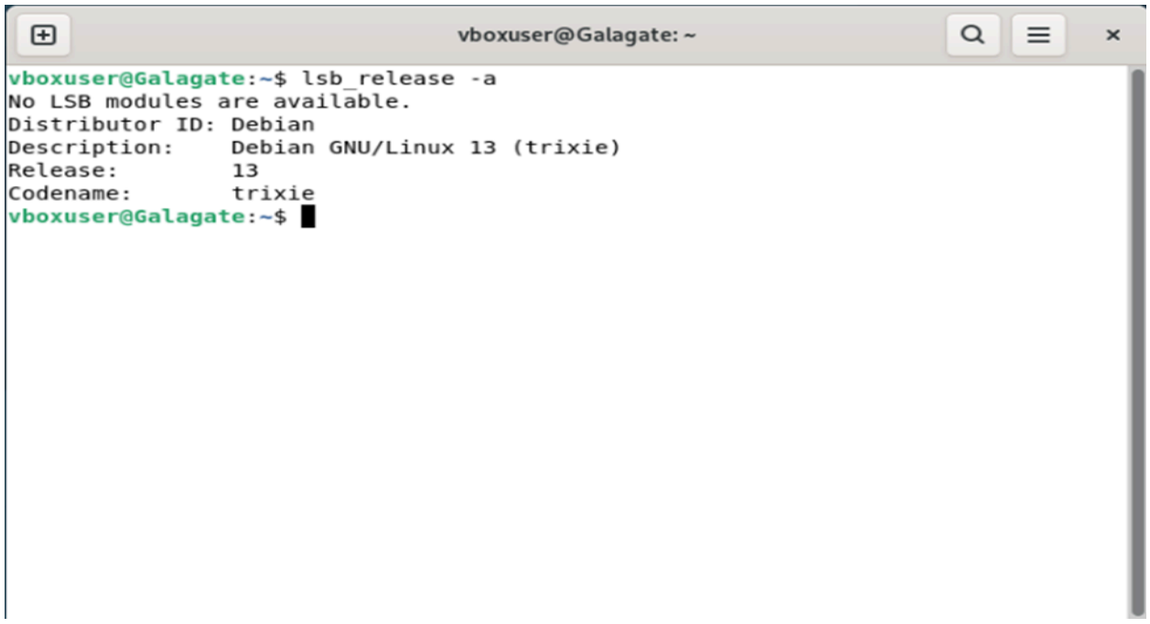
- After installation, Go to App then Open the terminal

2.1 Operating System Version

- Prompt: lsb_release -a

Purpose:

Displays detailed information about the installed Debian distribution.



2.2 Kernel Version

Prompt: uname -r

Purpose:

Shows the currently running Linux kernel version, which is essential for compatibility and security updates

.



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2.3 Shows Memory

Prompt: free -m

Purpose:

To show the system's **RAM and swap usage** in **megabytes**.

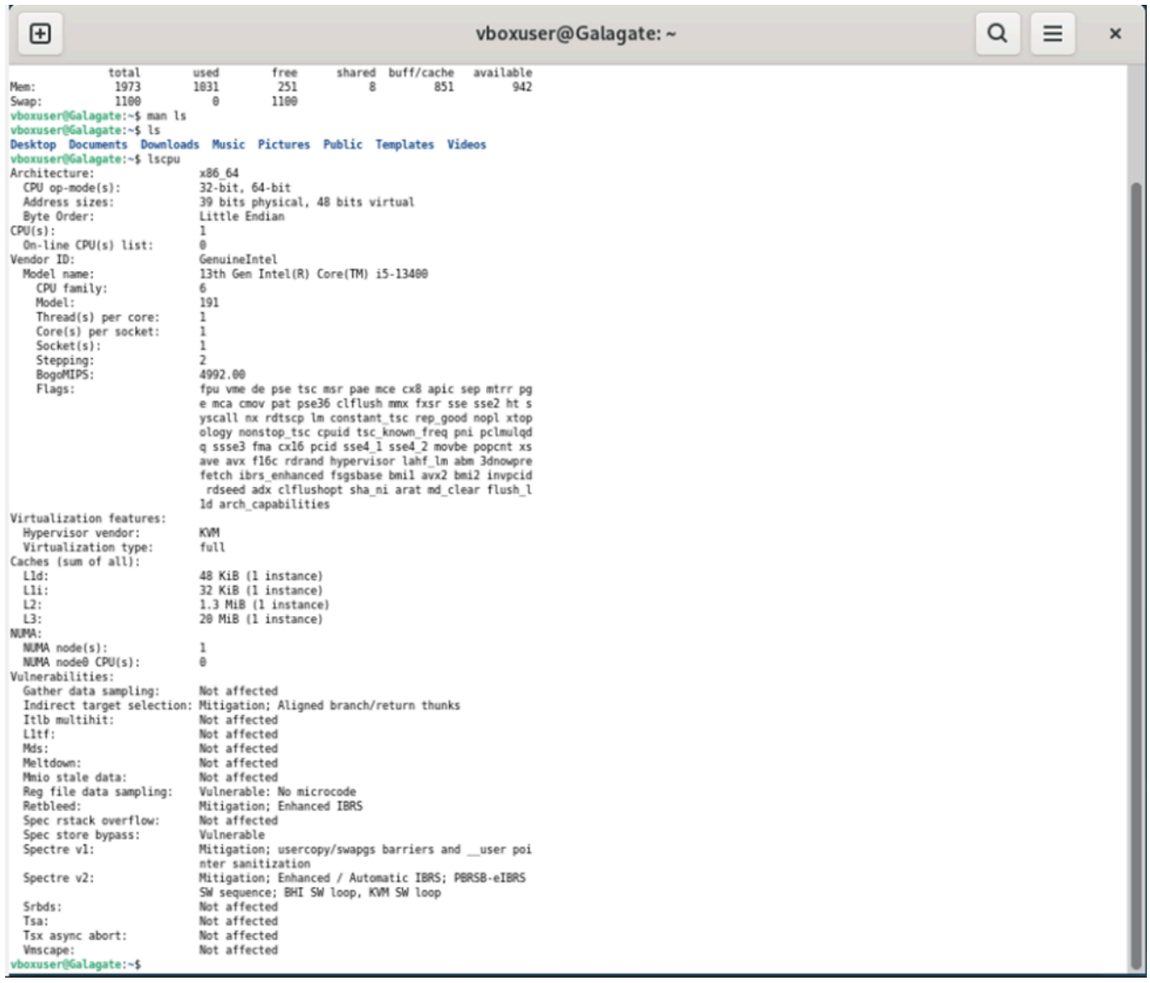
```
vboxuser@Galagate:~$ free -m
               total        used        free      shared  buff/cache   available
Mem:            1973        1031         251           8         851         942
Swap:           1100           0        1100
```

2.4 CPU Information

Prompt: lscpu

Purpose:

Provides details about the CPU architecture, number of cores, and processor model



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Essential Linux Administrative Commands

Documented Linux Commands (Using man)

3.0 mkdir – Make Directory

Prompt: man mkdir

Purpose:

To create new directories (folders) in the Linux file system, allowing users and system administrators to organize files and manage directory structures efficiently.



3.1 cp – Copy

Prompt: man cp

Purpose:

To copy files or directories from one location to another while keeping the original data intact, enabling file duplication, organization, and backup operations.



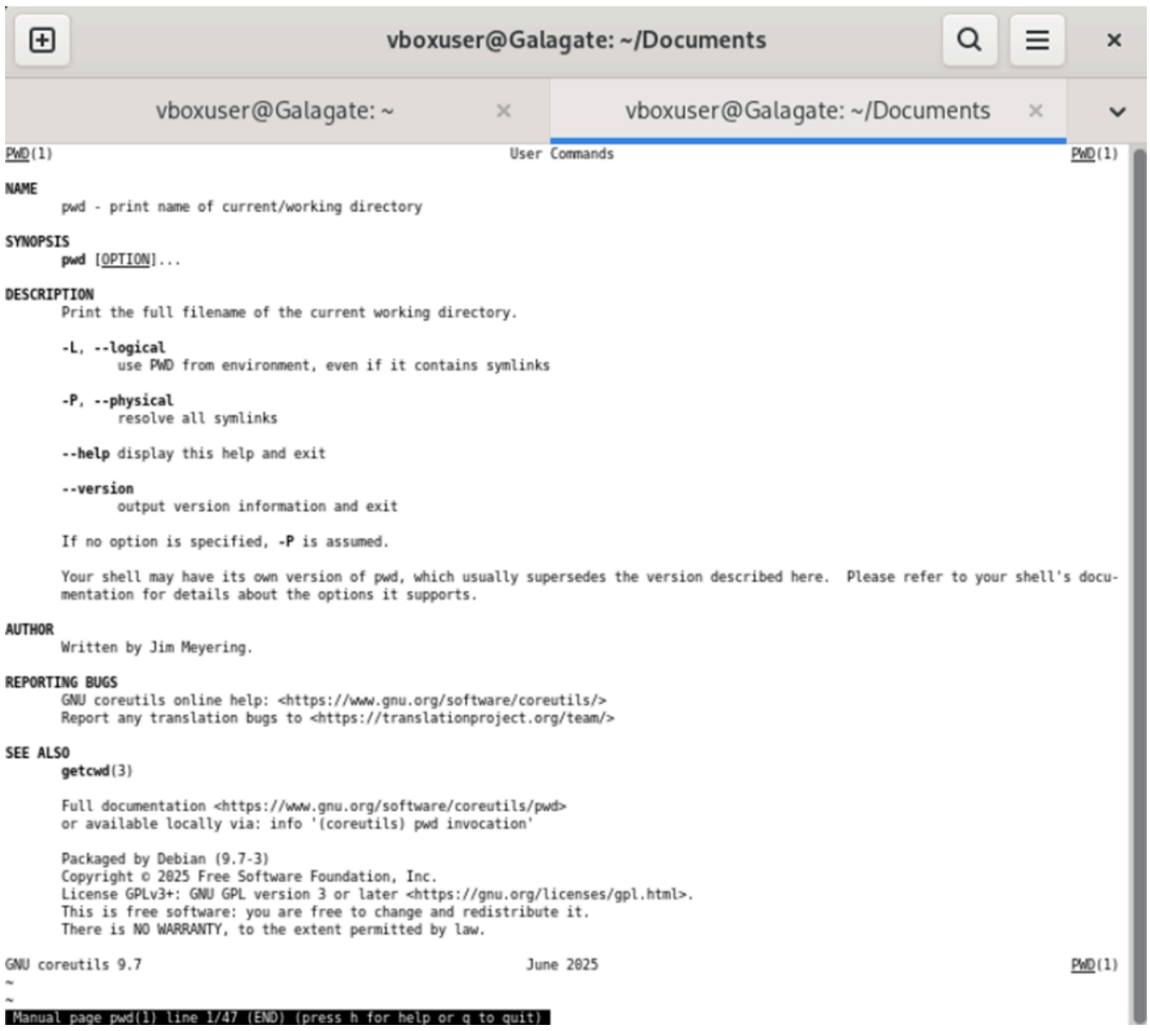
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3.2 pwd – Print Working Directory

Prompt: man pwd

Purpose:

To display the full absolute path of the current working directory, helping users and system administrators know their exact location within the Linux file system.



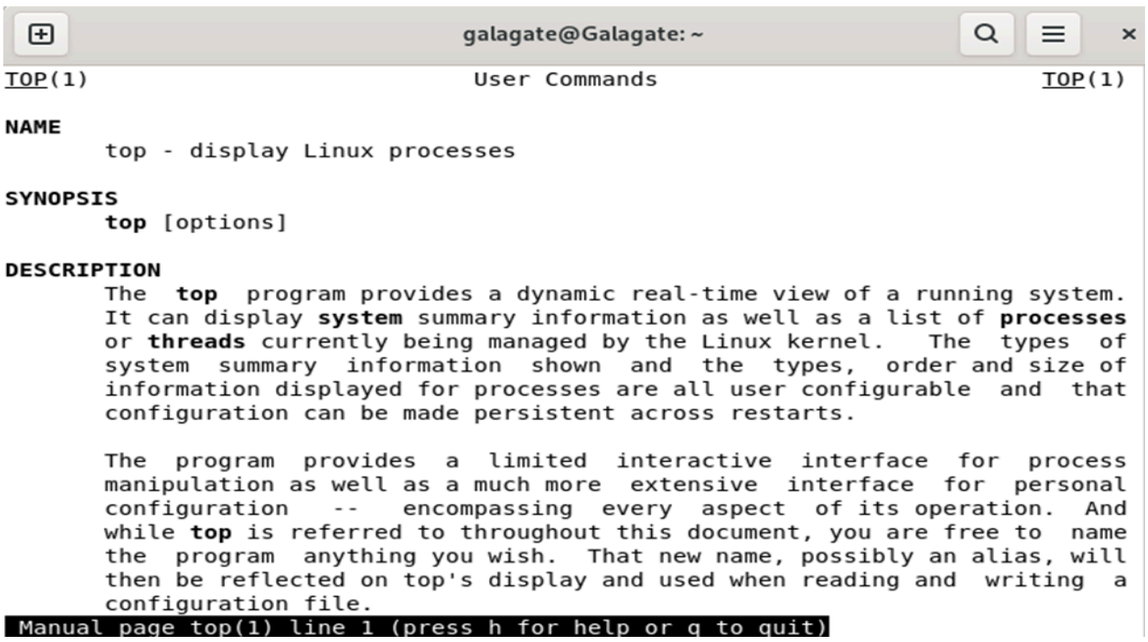
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3.3 top – Real-Time System Monitoring

Prompt: man top

Purpose:

- Displays real-time information about running processes and system resource usage.

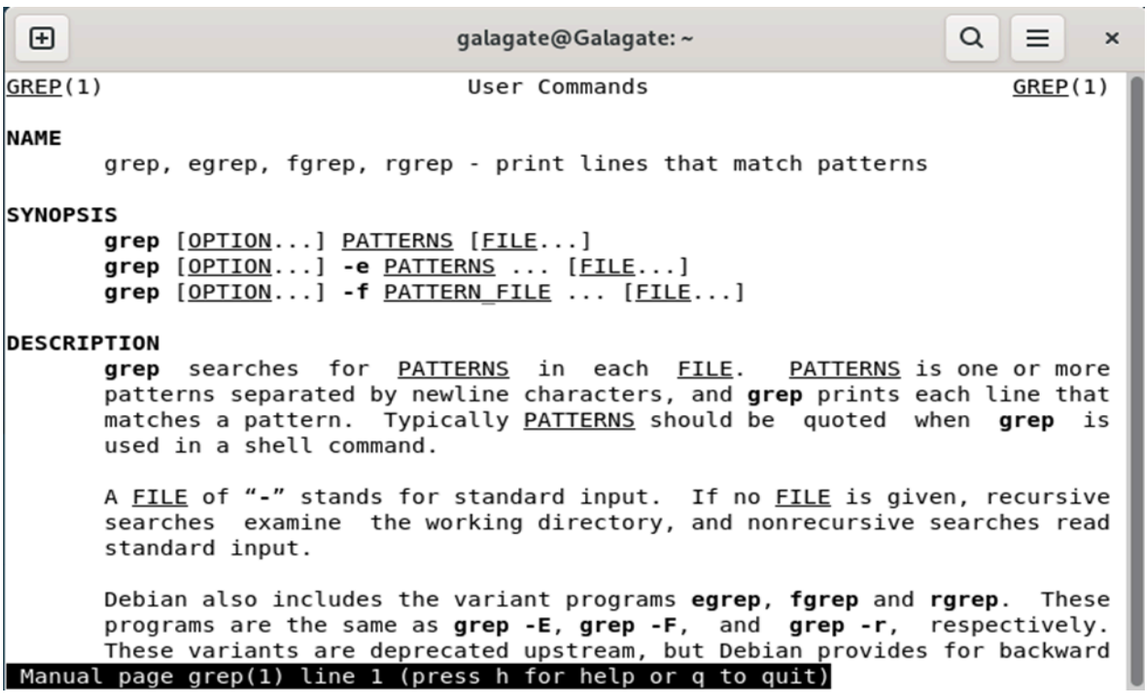


3.4 grep – Global Regular Expression Print.

Prompt: man grep

Purpose:

- Search for specific text or patterns in files or command output and display only the matching lines.



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User, Group, and File Permission Management in Linux

In this laboratory activity, I used root privileges to perform user, group, and file permission management in Linux. I use su – to switch to root with a full login environment Then enter the root password (set during Debian install).



1. Add a local user account to your Linux system using the “useradd” command that has a username of “nicole” , fullname of “Nicoljul Tesla”, “jay” fullname “Jay Abando” that uses /bin/sh as its default shell. Then set a unique password to each user

```
root@Galagate:~# useradd -m -c "Nicoljul Tesla" -s /bin/sh nicole
root@Galagate:~# passwd nicole
New password:
Retype new password:
passwd: password updated successfully
root@Galagate:~# useradd -m -c "Jay Abando" -s /bin/sh jay
root@Galagate:~# passwd jay
New password:
Retype new password:
passwd: password updated successfully
root@Galagate:~# ls /home
galagate  jay  nicole
```

- useradd

  creates new user

- -m

  create home directory

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- -c

  full name/comment field

- -s /bin/sh

  default shell

- passwd

  assign secure password

2. Assumed that Jay owned a file named “sampleFile.txt” and the group is MIS. The file permission of the owner is read access and group is (420) access

```
root@Galagate:~# groupadd MIS
root@Galagate:~# touch /home/jay/sampleFile.text
root@Galagate:~# ls /home/jay/
sampleFile.text
root@Galagate:~# chown jay:MIS /home/jay/sampleFile.text
root@Galagate:~# chmod 420 /home/jay/sampleFile.text
root@Galagate:~# ls -l /home/jay/sampleFile.text
-r---w---- 1 jay MIS 0 Feb 14 13:08 /home/jay/sampleFile.text
```

chown = Change Owner

- groupadd

  is used to create a new group on the system.

- home

  system directory containing all user home folders

- /home/jay/sampleFile.txt

  specific file inside Jay’s home directory, whose ownership is being changed

- Owner

  jay

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- Group

  MIS

- File affected

  sampleFile.txt

chmod = Change Mode (permissions)

It controls who can read, write, or execute a file.

- Owner = 4 → Read only
- Group = 2 → Write only o
- Others = 0 → No access

- ls

  list files

- -l

  long format: shows detailed information about each file or directory

Part	Value	Meaning
File Type	-	Regular file
Owner Permission	r--	Owner can read only
Group Permission	-w-	Group can write only
Others Permission	---	Others have no access
Links	1	Number of hard links to this file
Owner	jay	User who owns the file
Group	MIS	Group assigned to the file
Size	0	File size in bytes
Date	Feb 14	Last modification date
Time	11:46	Last modification time

3. Create also a group name “SmartCampus” which is a system account. And add “jay” to the “SmartCampus” group.

```
root@Galagate:~# groupadd -r SmartCampus
root@Galagate:~# usermod -aG SmartCampus jay
root@Galagate:~# getent group SmartCampus
SmartCampus:x:986:jay
root@Galagate:~# groups jay
jay : jay SmartCampus
root@Galagate:~# id jay
uid=1002(jay) gid=1002(jay) groups=1002(jay),986(SmartCampus)
```

• -r

system group

• usermod -aG

append user to group (without removing existing groups)

• getent group SmartCampus

Check if the group exists

• groups jay

Check jay’s group membership

• id jay

Detailed user info